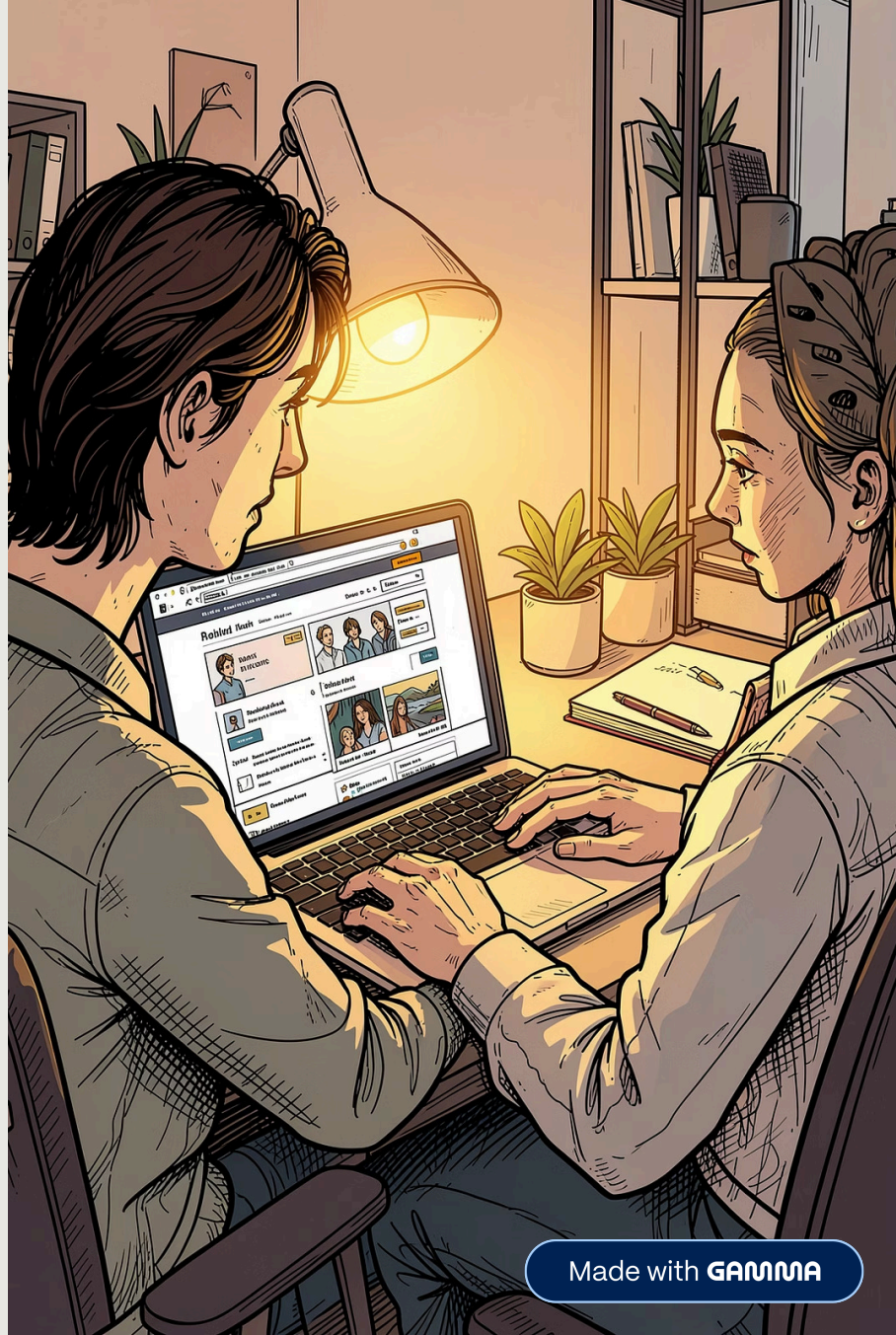
# ONLINE SHOPPERS INTENTION ANALYSIS

A comprehensive machine learning project using R to analyze customer browsing behavior, identify factors influencing online purchases, and predict whether a visitor will complete a transaction — generating actionable business insights for e-commerce optimization.

MACHINE LEARNING | R PROGRAMMING | E-COMMERCE ANALYTICS



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# INTRODUCTION & PROBLEM STATEMENT

## BACKGROUND

Online retailers continuously seek ways to increase customer conversions and maximize revenue. By analyzing user browsing patterns, businesses can identify behaviors associated with purchasing decisions and develop personalized marketing strategies. The rapid growth of e-commerce has generated vast amounts of customer interaction data — understanding this data is crucial for improving conversion rates and enhancing user experiences.

## THE CORE PROBLEM

Many visitors browse e-commerce websites without making purchases. Identifying potential buyers during browsing sessions enables organizations to target customers effectively and improve conversion rates. This project analyzes the Online Shoppers Intention dataset using R and machine learning techniques to address this challenge.

## PROJECT OBJECTIVES

- Analyze customer browsing behavior
- Identify factors affecting purchase decisions
- Develop predictive models for purchase intention
- Generate business recommendations to improve sales performance

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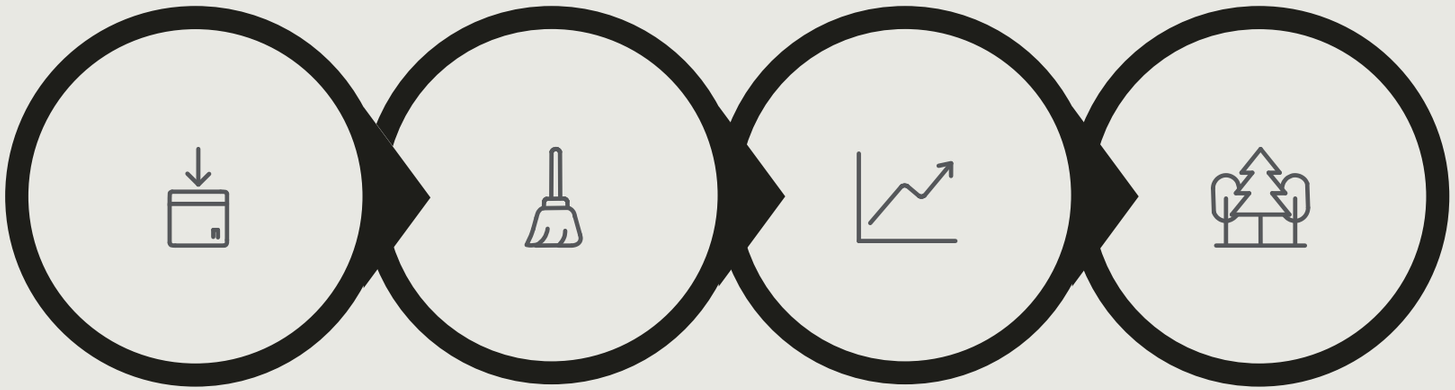
# DATASET DESCRIPTION

The Online Shoppers Intention dataset contains information about user sessions on an e-commerce website, comprising 18 features covering page interactions, visitor attributes, and session context. The target variable is **Revenue** — TRUE indicates a purchase was made, FALSE indicates no purchase.

| Feature | Description |
|---|---|
| Administrative / Duration | Number of administrative pages visited and time spent on them |
| Informational / Duration | Number of informational pages visited and time spent on them |
| ProductRelated / Duration | Number of product-related pages viewed and time spent on product pages |
| BounceRates / ExitRates | Percentage of visitors leaving after one page / percentage of exits from website pages |
| PageValues | Value assigned to visited pages |
| SpecialDay / Month | Closeness to a special shopping day; month of visit |
| OperatingSystems / Browser | Visitor operating system and browser type |
| Region / TrafficType | Geographic region and source of website traffic |
| VisitorType / Weekend | Returning or New Visitor; weekend session indicator |
| Revenue (Target) | TRUE = Customer made a purchase; FALSE = No purchase |

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# METHODOLOGY & DATA PREPROCESSING




The methodology follows a structured four-stage pipeline from raw data import through model development and evaluation.

## DATA COLLECTION

The dataset was imported into R Notebook using CSV file operations, providing a reproducible analytical environment. The following R packages were leveraged throughout the project: **Tidyverse** for data manipulation, **ggplot2** for visualization, and **Caret** for model training and evaluation.

## DATA CLEANING

- Missing value identification — no significant missing values were found
- Data type verification and conversion of categorical variables into factors
- Removal of inconsistencies
- Data normalization and scaling

## TRAIN-TEST SPLIT

The dataset was divided into an **80% training set** and a **20% testing set**, allowing for unbiased model evaluation. This split ensures that model performance metrics reflect genuine predictive capability on unseen data.

## SKILLS DEMONSTRATED

This project demonstrates proficiency in data cleaning, data wrangling, exploratory data analysis, statistical analysis, data visualization, predictive modeling, classification algorithms, feature importance analysis, business intelligence, and customer analytics.

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# EXPLORATORY DATA ANALYSIS

EDA was conducted to understand customer behavior patterns before modeling. Key findings across revenue distribution, visitor type, monthly trends, and correlation analysis informed feature selection and model strategy.

## REVENUE DISTRIBUTION

Analysis revealed that most visitors do not complete purchases, and the dataset exhibits **class imbalance**. However, conversion opportunities exist for optimization — understanding which sessions convert is the core modeling challenge.

## VISITOR TYPE ANALYSIS

Findings indicate that **returning visitors are more likely to generate revenue**, while new visitors have lower conversion rates. VisitorType emerged as a meaningful segment for targeted marketing strategies.

## MONTHLY TRENDS

Customer purchases fluctuate seasonally. Higher conversion activity is typically observed during **holiday shopping periods**, promotional campaigns, and peak retail seasons — confirming the importance of the SpecialDay and Month features.

## CORRELATION ANALYSIS

Strong **positive relationships** were observed between ProductRelated and ProductRelated_Duration, and between PageValues and Revenue. **Negative relationships** were identified between BounceRates and Revenue, and between ExitRates and Revenue — confirming that disengagement signals reduce purchase likelihood.

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# MACHINE LEARNING MODELS

## LOGISTIC REGRESSION

**Purpose:** Predict customer purchase probability using a statistical classification method.

**Advantages:**

- Easy interpretation of coefficients
- Fast computation
- Suitable as a baseline model

**Limitations:**

- Assumes linear relationships between features
- Less effective for capturing complex, nonlinear patterns in customer behavior

## RANDOM FOREST

**Purpose:** Build multiple decision trees and combine predictions for improved accuracy.

**Advantages:**

- Handles nonlinear relationships effectively
- Robust against overfitting through ensemble averaging
- Provides feature importance rankings for interpretability

**Expected Outcome:** Random Forest typically outperforms Logistic Regression in customer behavior prediction tasks due to its ability to model complex, nonlinear interactions within the data.

Both models were evaluated using a consistent 80/20 train-test split to ensure fair comparison of predictive performance across all evaluation metrics.

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# MODEL EVALUATION METRICS

Several metrics were used to evaluate model performance, each providing a different lens on predictive quality — especially important given the class imbalance in the dataset.

## ACCURACY

Measures overall prediction correctness. Formula: Correct Predictions ÷ Total Predictions. Provides a general sense of model performance but can be misleading with imbalanced classes.

## PRECISION

Measures the proportion of *predicted purchases* that were *actual purchases*. High precision means fewer false alarms — critical for targeting efficiency.

## RECALL

Measures the proportion of *actual purchasers* correctly identified by the model. High recall ensures fewer potential buyers are missed during targeting.

## F1 SCORE

Harmonic mean of Precision and Recall. Balances both metrics into a single score — ideal for comparing models when class imbalance is present.

## ROC-AUC

Evaluates model discrimination capability across all classification thresholds. Higher AUC values indicate stronger predictive performance and better separation of buyers from non-buyers.

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# FEATURE IMPORTANCE ANALYSIS

Random Forest feature importance analysis revealed the top five predictors of purchase intention, providing direct insight into which customer behaviors most strongly signal buying intent.

## 1 PAGEVALUES

The **most influential feature** overall. Higher page values strongly correlate with purchase intent, making it the single strongest predictor of whether a visitor will convert.

## 2 EXITRATES

High exit rates **reduce purchase probability** significantly. Visitors who frequently exit pages are demonstrating disengagement signals that lower conversion likelihood.

## 3 PRODUCTRELATED_DURATION

**Longer engagement** with product pages increases conversion likelihood. Time spent on product pages is a strong behavioral indicator of genuine purchase consideration.

## 4 BOUNCERATES

Higher bounce rates indicate **lower customer interest** and reduced purchase probability. Single-page sessions rarely result in conversions.

## 5 PRODUCTRELATED

Viewing **more product pages** reflects stronger buying intent. The number of product-related pages visited is a clear signal of active shopping behavior.

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# BUSINESS INSIGHTS & RECOMMENDATIONS

## CUSTOMER BEHAVIOR INSIGHTS

**High-Converting Visitors** are characterized by: returning customer status, longer browsing sessions, higher page values, and more product interactions. These are the sessions most likely to result in revenue.

**Low-Converting Visitors** exhibit: high bounce rates, high exit rates, and minimal product engagement. These sessions represent the largest opportunity for intervention and optimization.

## STRATEGIC RECOMMENDATIONS

- **Improve Product Pages:** Enhance product descriptions, images, and customer reviews to increase user engagement and improve conversion rates.
- **Personalized Marketing:** Implement recommendation systems, personalized offers, and dynamic promotions for higher customer satisfaction and increased purchases.
- **Retargeting Campaigns:** Target users who viewed products extensively, abandoned sessions, or added products without purchasing — recovering lost sales opportunities.
- **Customer Retention Programs:** Develop loyalty programs, membership rewards, and exclusive discounts to increase repeat purchases and higher customer lifetime value.

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# CONCLUSION & FUTURE DIRECTIONS

## PROJECT SUMMARY

This project successfully analyzed online shopper behavior and developed predictive models for purchase intention. Exploratory analysis identified customer engagement metrics as major indicators of purchasing decisions. **Random Forest demonstrated strong predictive capability** due to its ability to model complex relationships within customer behavior data, outperforming the Logistic Regression baseline.

The findings provide valuable insights for e-commerce businesses seeking to optimize conversion rates, improve marketing effectiveness, and enhance customer experiences.

## TECHNOLOGIES USED

R Programming, R Notebook, Tidyverse, ggplot2, Caret, Random Forest, Logistic Regression, Correlation Analysis, Machine Learning, Data Visualization

## FUTURE ENHANCEMENTS

Future work may include the following improvements to further enhance predictive performance and business value:

### XGBOOST & GRADIENT BOOSTING

Advanced ensemble methods for improved accuracy

### HYPERPARAMETER TUNING

Systematic optimization of model parameters

### DEEP LEARNING APPROACHES

Neural network models for complex pattern recognition

### REAL-TIME PREDICTION SYSTEMS

Live purchase intention scoring during browsing sessions

### CUSTOMER SEGMENTATION

Clustering algorithms to identify distinct visitor profiles

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